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Simple Payback & Rate of Return

* Time required for an investment to generate sufficient cash flows to recover the initial investments.

$$SP = \frac{\text{Initial cost}}{\text{Saving/year}}$$

Q1 Calculate the simple payback of a lighting retrofit that will cost \$1000 to implement & will save \$250 per year.

$$SP = \frac{1000}{250} = 4 \text{ years}$$

ROR = It represents the annual return on the investment, it is the reciprocal of simple payback.

$$ROR = \frac{\text{Saving/year}}{\text{Initial cost}} = \frac{250}{1000} = 0.25 \text{ or } 25\%$$

* Payback Period with uneven cash flow;

Payback = years before full recovery + $\frac{\text{unrecovered cost at start of the year}}{\text{cash flow during the year}}$

Q Initial Investment: Rs 50000

It generates a cash flow of Rs 15000, Rs 20000, Rs 14000, Rs 16000 during the next 4 years. Find out the payback period.

Year Cash flow

0 -50000

1 -35000 15000

2 -15000 20000

3 -10000 14000

4 15000 16000

$$\therefore \text{Payback} = 3 + \frac{1000}{16000}$$

$$= 3.0625 \text{ yrs}$$

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Time value of Money

* Entrepreneurs & growing companies are interested in acquiring money today to make a profit with it tomorrow.

Thus, in the right hands \$100 today is worth more than \$100 tomorrow; money has a time value component.

* To compare investment options with cash flows that occur at different times, it is useful to convert all cash flows to a common time.

The most common way to do so is to convert all cash flow into their 'present value'.

Net 'Present value'

or Present value of a future Amount:

* Present value of future amount:

Consider Present amount P , rate of interest i over n compounding years. The future amount F can be determined as

$$F_0 = 100$$

$$F_1 = 100 + 100(0.05) = P + P(i) = P(1+i)$$

$$F_2 = [100 + 100(0.05)] + [100 + 100(0.05)](0.05)$$

$$= P(1+i) + P(1+i)i$$

$$= P(1+i)(1+i) = P(1+i)^2$$

$$F_n = P(1+i)^n$$



$$F = P(1+i)^n$$

This can be rearranged to show the present value of a future amount;

$$P = \frac{F}{(1+i)^n}$$

The factor $(1+i)^{-n}$ or $\frac{1}{(1+i)^n}$ is sometimes called the present worth factor, PWF.

Q Someone promises to pay you \$1000 in 5 years. If the interest rate is 10% per year, what amount would you take today that is equivalent to \$1000 in 5 years.

$$P = \frac{F}{(1+i)^n} = \frac{1000}{(1+0.1)^5} = \$621$$

* NPV = Discounted cash flow valuation

NPV > 0 → Accept Project

NPV < 0 → Reject Project

NPV = 0 → May accept or reject

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ECONOMICS

* Present value of series of cash flows

P_n = present value of n payments of amount A

$$P_0 = 0$$

$$P_1 = \frac{A}{(1+i)}$$

$$P_2 = \frac{A}{(1+i)^2} + \frac{A}{(1+i)}$$

$$x = \frac{1}{(1+i)}$$

$$P_n = A \left[\frac{1}{(1+i)^n} + \frac{1}{(1+i)^{n-1}} + \dots + \frac{1}{1+i} \right]$$

$$a = \frac{A}{(1+i)}$$

$$P_n = A \left[\frac{1 - (1+i)^{-n}}{i} \right] \quad \text{using Geom. Prog.} \Rightarrow a \left[\frac{1 - x^n}{1 - x} \right]$$

* A standard efficiency furnace costs \$100 & consumes \$40 per year in fuel over 10 year lifetime. A high efficiency furnace costs \$200 & consumes \$20 per year in fuel over its 10 year lifetime. If interest rate are 10% per year, which is better investment.

$$P_{SL} = \$100 + \$40 \left[\frac{1 - (1+0.1)^{-10}}{0.1} \right] = \$346$$

$$P_{HL} = \$200 + \$20 \left[\frac{1 - (1+0.1)^{-10}}{0.1} \right] = \$323$$

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$$\text{Prigh saving} = \$346 - \$323 = \$23$$

are directly

$$\begin{aligned} \text{Prigh saving} &= -\$100 + 20 \left[\frac{1 - (1 + 0.1)^{-10}}{0.10} \right] \\ &= \$23 \end{aligned}$$

* Effect of discount rate

- The discount rate is the expected rate of return from alternative investment
- High discount rates reflect the belief that a large profit can be made from an alternative investment.

→ In the previous problem if the discount rate is considered 30% instead of 10%;

$$\begin{aligned} \text{Prigh saving} &= -100 + 20 \left[\frac{1 - (1 + 0.3)^{-10}}{0.30} \right] \\ &= -38 \$ \end{aligned}$$

* It indicates that at high discount rate, the standard furnace with low initial cost is the better investment.

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Using ^{method}
* Calculate the payback period & NPV for the following Projects & choose the project to be executed
Consider the discount rate 10%.

Project A (\$)	Project B (\$)	Year
-1200	-1200	0
400	400	1
400	400	2
400	350	3
200	800	4
200	800	5

A

Payback period = 3 years

B Payback period = $3 + \frac{50}{800}$

= 3.0625 years

$$\begin{aligned} \text{NPV A} &= -1200 + \frac{400}{(1+0.1)} + \frac{400}{(1+0.1)^2} + \frac{400}{(1+0.1)^3} + \frac{200}{(1+0.1)^4} \\ &\quad + \frac{200}{(1+0.1)^5} \\ &= \$ 55 \end{aligned}$$

$$\begin{aligned} \text{NPV B} &= -1200 + \frac{400}{(1+0.1)} + \frac{400}{(1+0.1)^2} + \frac{350}{(1+0.1)^3} + \frac{800}{(1+0.1)^4} + \frac{800}{(1+0.1)^5} \\ &= \$ 800.32 \end{aligned}$$

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Practice Problem

* Consider investing in a real estate project.

Purchase of land Rs 200000

Project will generate a cash flow of Rs 30000/year for next 8 years

Cost of capital is 12%

Calculate the NPV & determine whether it is financially viable or not.

Solⁿ
$$P_n = A \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

$$P_n = 30000 \left[\frac{1 - (1+0.12)^{-8}}{0.12} \right]$$

$$= 149029 \text{ Rs}$$

$$NPV = -200000 + 149029$$

$$= -50970 \text{ Rs}$$

$NPV < 0$; The project is not viable financially

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* Calculate the NPV if you invest \$2000 now,
receive 3 yearly payments of \$100 each, plus
\$2500 in the 3rd year. Using 10% discount rate.

$$= PV_0 = -\$2000$$

$$\text{Year 1} = PV = \frac{\$100}{1.1} = \$90.91$$

$$\text{Year 2} = PV = \frac{\$100}{(1.1)^2} = \$82.64$$

$$\text{Year 3} = PV = \frac{\$100}{(1.10)^3} = \$75.13$$

$$\text{Year 3 (Final Payment)} PV = \frac{\$2500}{(1.10)^3} = \$1878.29$$

$$\begin{aligned} NPV &= -2000 + 90.91 + 82.64 + 75.13 + 1878.29 \\ &= \$126.97 \end{aligned}$$

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9 Initial investment = 150000

Cash flow each year = 50000

Discount rate = 10%

No. of year (n) = 5

$$NPV = \sum_{t=1}^n \frac{F}{(1+i)^t} - F_0$$

$$= \frac{50000}{(1+0.1)^1} + \frac{50000}{(1+0.1)^2} + \frac{50000}{(1+0.1)^3} + \frac{50000}{(1+0.1)^4} + \frac{50000}{(1+0.1)^5} - 150000$$

$$= \frac{50000}{(1+0.1)^1} + \frac{50000}{(1+0.1)^2} + \frac{50000}{(1+0.1)^3} + \frac{50000}{(1+0.1)^4} + \frac{50000}{(1+0.1)^5} - F_0$$

$$= 189539.34 - 150000$$

$$= \text{Rs. } 39540$$

* IRR (Internal Rate of Return)

The discount rate which sets the NPV of all cash flow equal to 0.

* It helps to determine the yield on an investment

$$NPV = 0 = -\text{Initial Investment} + \frac{\text{Cash Flow}_1}{(1+IRR)^1} + \dots + \frac{CF_n}{(1+IRR)^n}$$

* In excel (ROI) is called (IRR)

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IRR*

A company is deciding whether to purchase new equipment that costs \$500,000. Management estimates the life of the new asset to be four years & expects it to generate an additional \$160,000 of annual profit. In the fifth year, the company plans to sell the equipment for its salvage value of \$50,000.

Calculate IRR.

$$\text{For IRR } NPV = 0 = \text{Initial investment} + \frac{\text{Cash Flow}}{(1+IRR)^1} + \dots + \frac{CF_n}{(1+IRR)^n}$$

$$= -500000 + \frac{160000}{(1+IRR)^1} + \frac{160000}{(1+IRR)^2} + \frac{160000}{(1+IRR)^3} + \frac{160000}{(1+IRR)^4} + \frac{50000}{(1+IRR)^5}$$

$$X = 15.24\% \quad 13.27\%$$

IRR doesn't give you the return on the initial investment in terms of money.



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IRR = it assumes all the time period in a series of equal cash flows

XIRR = Gives the flexibility to assign specific dates to each individual cash flow, making it a much more accurate calculation.

8 The cash flow associated with a machine are

Initial cost = \$50000 & \$15000 additional profits for 5 years.

Determine the economic acceptability if i is a) 10% b) 15% c) 20%.

Also find the IRR of the project.

$$a) NPV_{i=10\%} = -50000 + 15000 \left[\frac{1 - (1+0.1)^{-5}}{0.1} \right] = \$6861.80$$

$$b) NPV_{i=15\%} = -50000 + 15000 \left[\frac{1 - (1+0.15)^{-5}}{0.15} \right] = \$282.33$$

$$c) NPV_{i=20\%} = -50000 + 15000 \left[\frac{1 - (1+0.2)^{-5}}{0.2} \right] = \$-5140.82$$

For IRR ; NPV will be 0 between 15% & 20%.

using interpolation

$$i = i_1 + \frac{i_2 - i_1}{y_2 - y_1} (y - y_1)$$

i = interest rate

y = NPV

$$= 0.15 + \frac{0.2 - 0.15}{(-5140.82 - 282.33)} (0 - 282.33)$$

$$= 0.1526 \text{ or } 15.26\%$$